

Ayan Syed

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WORK EXPERIENCE

Cobot - "Collaborative Robotics"

Santa Clara, CA

Robotics Software Intern

July 2025 - December 2025

- Accelerated robot manufacturing timeline by 2X by developing a foundational Python and C++ subassembly and bringup testing infrastructure using OpenHTF and TofuPilot, and developing APIs to automate robotic testing functionality
- Wrote Python ROS2 Visualization nodes to interface with Unity Frontend for intelligent dashboard control over the deployed fleet
- Enhanced reliability by debugging 100+ issues in critical ROS2 software and firmware, along with testing software and build systems in Bazel and ROS2 distributed communication platform, over Docker images on Linux, and NVIDIA Jetson Orin Hardware

NASA

AMES Research Center - Mountain View, CA

Research Intern

June 2024 - August 2024

- Extracted 21+ market KPIs and growth areas for an aviation research study with NASA Aviation data engineers by writing Numpy and Matplotlib Python scripts to analyze and evaluate 500GB+ of open source emission data
- Co-authored an alternative fuel technical paper summarizing 10 weeks of development and data findings; Published on N.T.R.S.

PROJECTS

iOS ARKit ROS2 Bridge

- Wrote an open-source Swift App to bridge iPhone ARKit into a plug-and-play RGB-D, LiDAR, and VIO sensors for ROS 2 robots
- Integrated RTAB-map VSLAM to enable localization, position precision, and loop closure - achieved <2cm SLAM precision
- Using Nav2 and Gazebo Simulation stack to enable real-time SLAM, autonomous navigation, and control on a mobile robot

EEG Machine Learning Neurofeedback Study Assistant

- Enhanced ADHD and student study focus rates by 22% by leading a 5-member independent research team developing a real-time brain-wave monitoring system utilizing OpenBCI for the 36 hours of De Anza Hacks 3.5 Hackathon
- Achieved 85% classification accuracy for concentration levels by developing a system with a multithreaded data filtering pipeline (Bandpass, Low/High Pass Filters, Fast Fourier Transform) and training 2 Scikit-learn Neural Networks
- Delivered personalized study insights and pacing to users by integrating Gemini AI on a ReactJS Web dashboard

Google Gemini Agentic Robotic Arm

- Launched an open-source 5DOF Desktop Robotic Arm Platform using Python and C++, used by 3+ College Research Teams
- Implemented a planar homography algorithm in OpenCV to find, classify, and locate objects for manipulation using April Tags
- Trained a Gemini Agent for intelligent control with PID/Motion Profiling, allowing users to manipulate objects with voice commands

CometKitz Non-Profit

- Inspired 200+ SCUSD elementary students by co-founding an educational non-profit program developing robotics kits 27% less expensive than the cheapest commercial option, using interactive web lessons, custom-designed PCB, and Arduino C++ firmware
- Led 18 software-specialized classes introducing robotic software to grades 4-8 by custom-building a full-stack Blockly web application that translated drag and drop block functions to sequential Arduino C++ code and complied with embedded firmware
- Secured \$10K+ in R&D/execution funding through key partnerships with Silicon Valley industry. Featured on NBC Bay Area News

SKILLS

Languages: Python, C++, Java, MATLAB, SQL, MASM, React, Bash, Swift, Unity C#

Tools: Linux/Ubuntu, Git, Github, JSON, AWS, RAG, CI/CD, Docker, OpenCV, NumPy, Sci-Kit Learn, Jira, Embedded Systems, Isaac Sim

EDUCATION

De Anza College

Expected Graduation: June 2028

Bachelor of Science in Computer Science and Engineering

4.0 GPA

- Data Structures & Algorithms, C++ & Python Series, Java, O.O.P., x86 Assembly Language, Computer Architecture, Cloud Compute

LEADERSHIP

DA Competitive Machine Learning

2025 Club President & LeadDeveloper

- Led top 100 World Ranking team developing novel high-performance Reinforcement Learning racing models on AWS DeepRacer

FTC Team 8872 Robopocalypse

2024 Team Captain & 2023 Software Lead

- Captained 15 person team to the top 1% in NorCal Region and qualified & competed at the 2024 Houston World Championship
- Recognized 2x by Santa Clara Mayor for exceptional team performance, leadership, and strong community impact efforts